



**Progressive Education Society's
Modern College of Engineering, Shivajinagar, Pune-05.**
(An Autonomous Institute Affiliated to Savitribai Phule Pune University)
Department of Computer Engineering

Lab Assignment No: 1

Title: Code Converter

Objective: To learn various code & its conversion

Problem Statement: To Design and implement the circuit for the following 4-bit Code conversion.

- i) Binary to Gray Code
- ii) Gray to Binary Code

Hardware & software requirements:

Digital Trainer Kit, IC 7404, IC 7432, IC 7408, IC 7486, Patch Cord, + 5V Power Supply

Theory:

There is a wide variety of binary codes used in digital systems. Some of these codes are binary- coded-decimal (BCD), Excess-3, Gray, octal, hexadecimal, etc. Often it is required to convert from one code to another. For example the input to a digital system may be in natural BCD and output may be 7-segment LEDs. The digital system used may be capable of processing the data in straight binary format. Therefore, the data has to be converted from one type of code to another type for different purpose. The various code converters can be designed using gates.

1) Binary Code:

It is straight binary code. The binary number system (with base 2) represents values using two symbols, typically 0 and 1. Computers call these bits as either off (0) or on (1). The binary code are made up of only zeros and ones, and used in computers to stand for letters and digits. It is used to represent numbers using natural or straight binary form.

It is a weighted code since a weight is assigned to every position. Various arithmetic operations can be performed in this form. Binary code is weighted and sequential code.

2) Gray Code:

It is a modified binary code in which a decimal number is represented in binary form in such a way that each Gray- Code number differs from the preceding and the succeeding number by a single bit. (E.g. for decimal number 5 the equivalent Gray code is 0111 and for 6 it is 0101. These two codes differ by only one bit position i. e. third from the left.) Whereas by using binary code there is a possibility of change of

all bits if we move from one number to other in sequence (e.g. binary code for 7 is 0111 and for 8 it is 1000). Therefore it is more useful to use Gray code in some applications than binary code.

The Gray code is a nonweighted code i.e. there are no specific weights assigned to the bit positions.

Like binary numbers, the Gray code can have any no. of bits. It is also known as reflected code.

Applications:

1. Important feature of Gray code is it exhibits only a single bit change from one code word to the next in sequence. This property is important in many applications such as Shaft encoders where error susceptibility increases with number of bit changes between adjacent numbers in sequence.
2. It is sometimes convenient to use the Gray code to represent the digital data converted from the analog data (Outputs of ADC).
3. Gray codes are used in angle-measuring devices in preference to straight forward binary encoding.
4. Gray codes are widely used in K-map

The disadvantage of Gray code is that it is not good for arithmetic operation

Binary to Gray Conversion

In this conversion, the input straight binary number can easily be converted to its Gray code equivalent.

1. Record the most significant bit as it is.
2. EX-OR this bit to the next position bit, record the resultant bit.
3. Record successive EX-ORed bits until completed.
4. Convert 0011 binary to Gray.

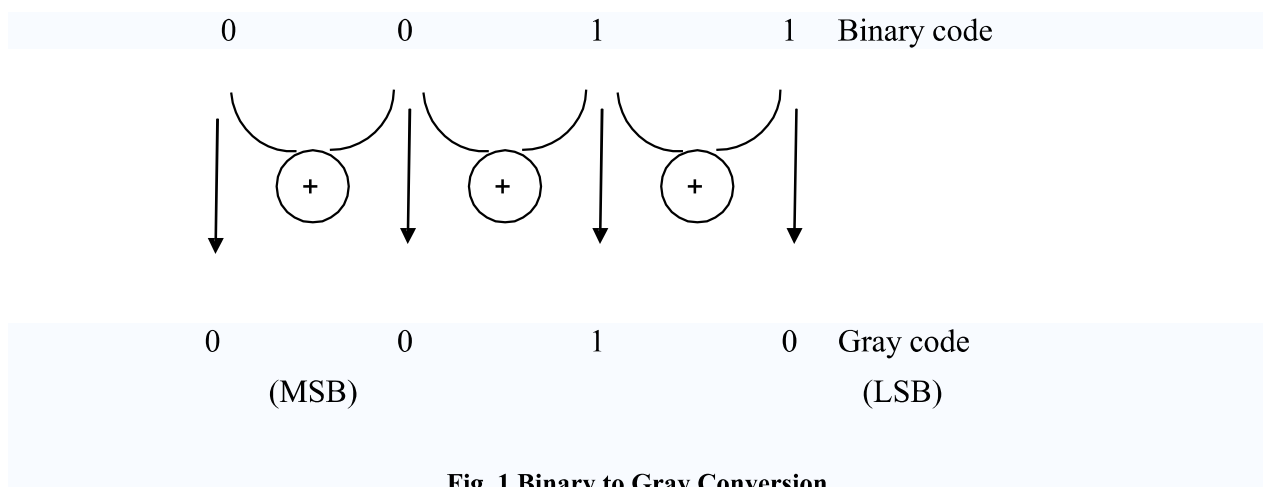
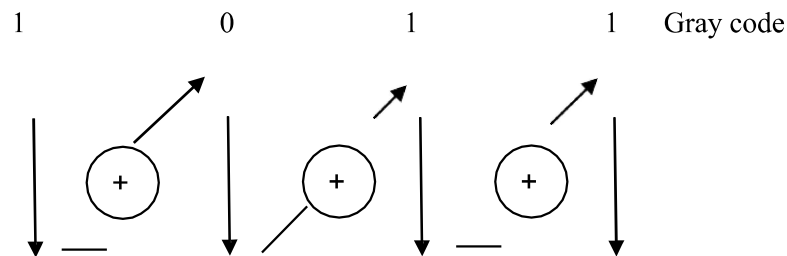


Fig. 1 Binary to Gray Conversion

Gray to Binary Conversion

1. The Gray code can be converted to binary by a reverse process.
2. Record the most significant bit as it is.
3. EX-OR binary MSB to the next bit of Gray code and record the resultant bit.
4. Continue the process until the LSB is recorded.
5. Convert 1011 Gray to Binary code.



1	1	0	1	Binary code
(MSB)			(LSB)	

Fig. 2 Gray to Binary Conversion

Design:

A) Binary to Gray Code Conversion:

1) Truth Table:

Table 1 Binary to Gray Code Conversion

INPUT (BINARY CODE)				OUTPUT (GRAY CODE)			
B ₃	B ₂	B ₁	B ₀	G ₃	G ₂	G ₁	G ₀
0	0	0	0	0	0	0	0



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0	0	0	1	0	0	0	1
0	0	1	0	0	0	1	1
0	0	1	1	0	0	1	0
0	1	0	0	0	1	1	0
0	1	0	1	0	1	1	1
0	1	1	0	0	1	0	1
0	1	1	1	0	1	0	0
1	0	0	0	1	1	0	0
1	0	0	1	1	1	0	1
1	0	1	0	1	1	1	1
1	0	1	1	1	1	1	0
1	1	0	0	1	0	1	0
1	1	0	1	1	0	1	1
1	1	1	0	1	0	0	1
1	1	1	1	1	0	0	0

2) K-Map for Reduced Boolean Expressions of Each Output:

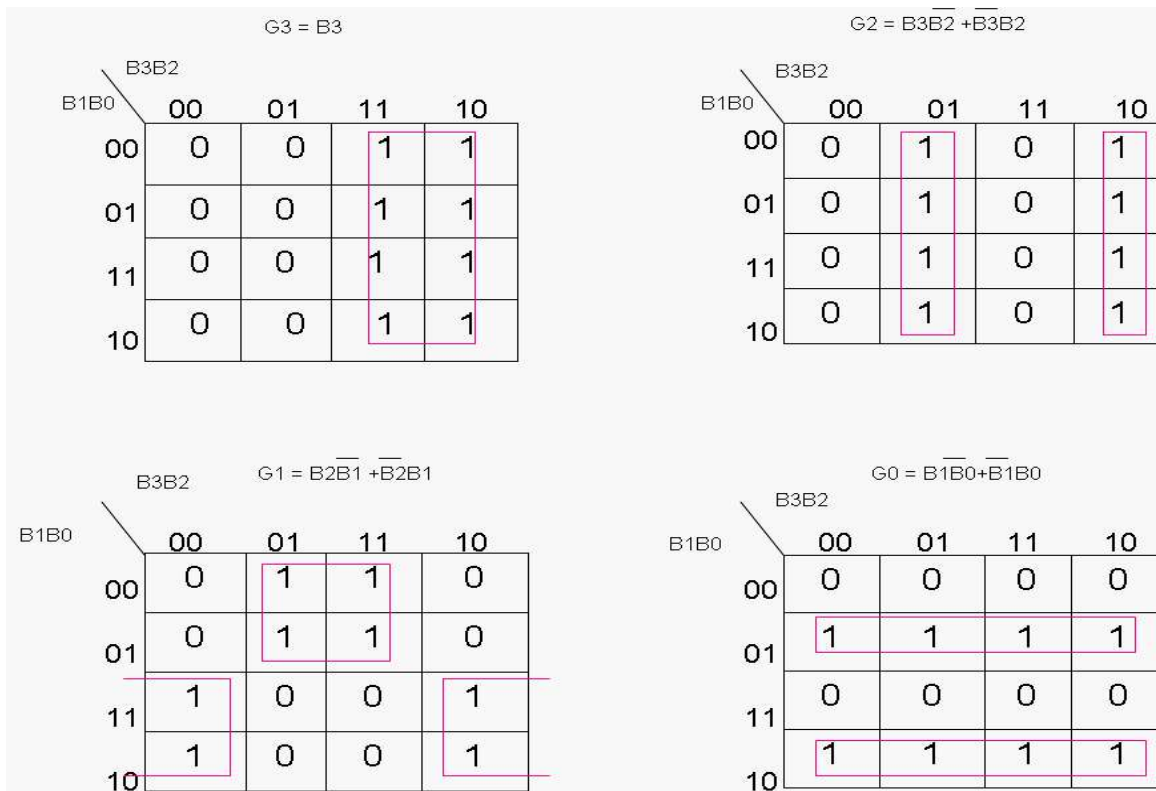
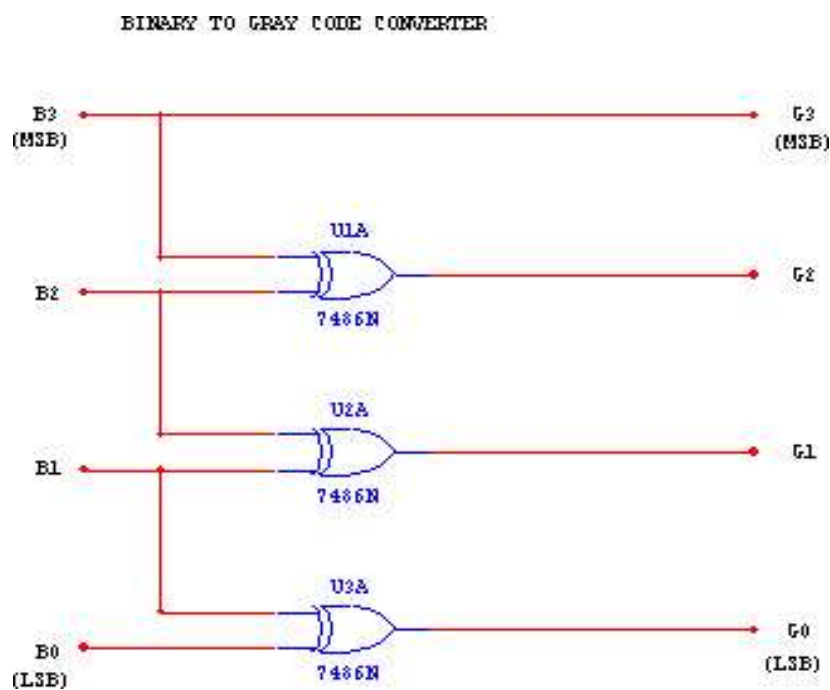


Fig. 4 K-Map for Reduced Boolean Expressions of Each Output (Gray Code)

3) Circuit Diagram:





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Fig. 5 Logical Circuit Diagram for Binary to Gray Code Conversion

B) Gray to Binary Code Conversion:

1) Truth Table:

Table 2 Gray to Binary Code Conversion

INPUT (GRAY CODE)				OUTPUT (BINARY CODE)			
G_3	G_2	G_1	G_0	B_3	B_2	B_1	B_0
0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	1
0	0	1	1	0	0	1	0
0	0	1	0	0	0	1	1
0	1	1	0	0	1	0	0
0	1	1	1	0	1	0	1
0	1	0	1	0	1	1	0
0	1	0	0	0	1	1	1
1	1	0	0	1	0	0	0
1	1	0	1	1	0	0	1
1	1	1	1	1	0	1	0
1	1	1	0	1	0	1	1
1	0	1	0	1	1	0	0
1	0	1	1	1	1	0	1
1	0	0	1	1	1	1	0
1	0	0	0	1	1	1	1

2) K-Map for Reduced Boolean Expressions of Each Output:

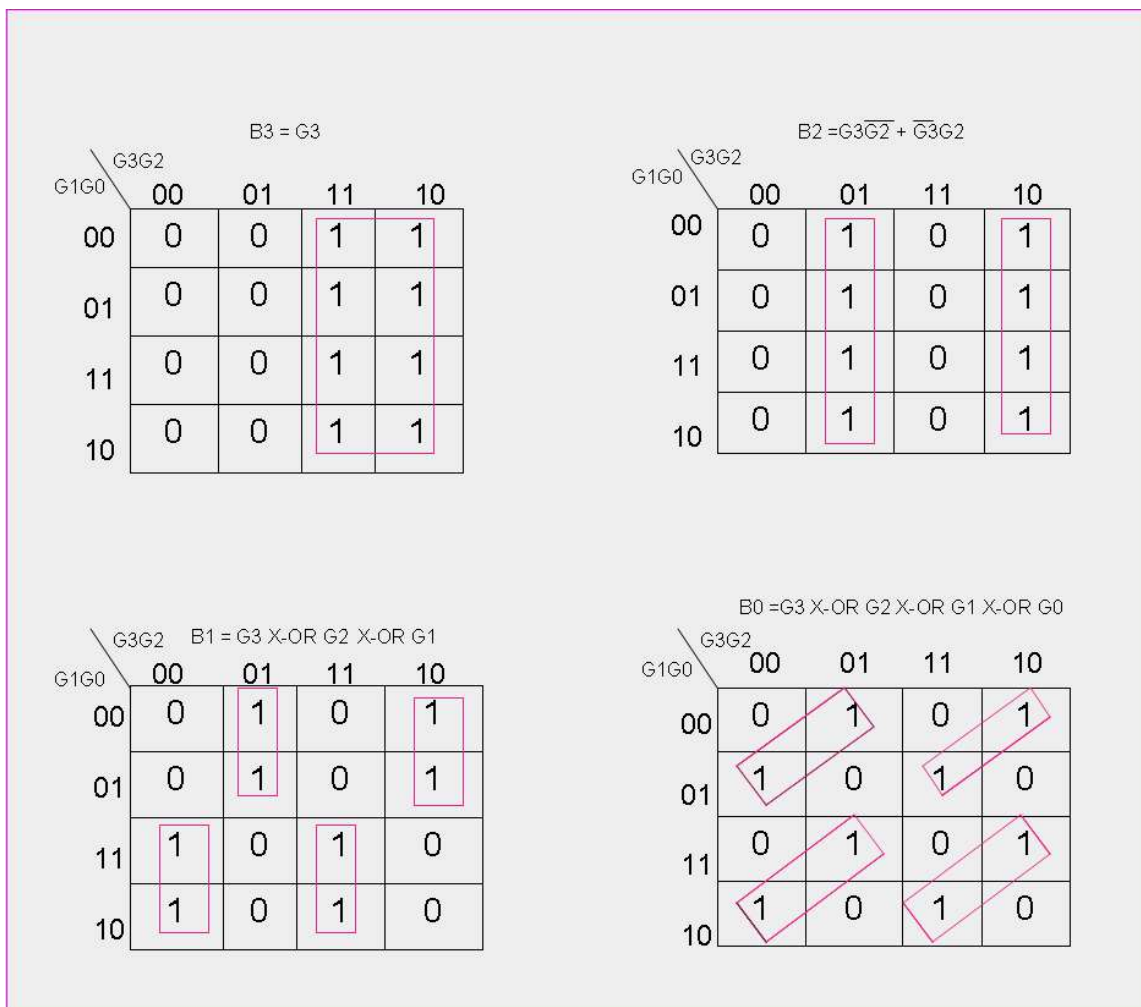


Fig. 6 K-Map for Reduced Boolean Expressions of Each Output (Binary Code)

00	0	1	0	1
01	1	0	1	0
11	0	1	0	1
10	1	0	1	0

Note:-Use this k-map instead one that is given above.

$$B0 = G3 \text{ X-OR } G2 \text{ X-OR } G1 \text{ X-OR } G0$$

3) Circuit Diagram:

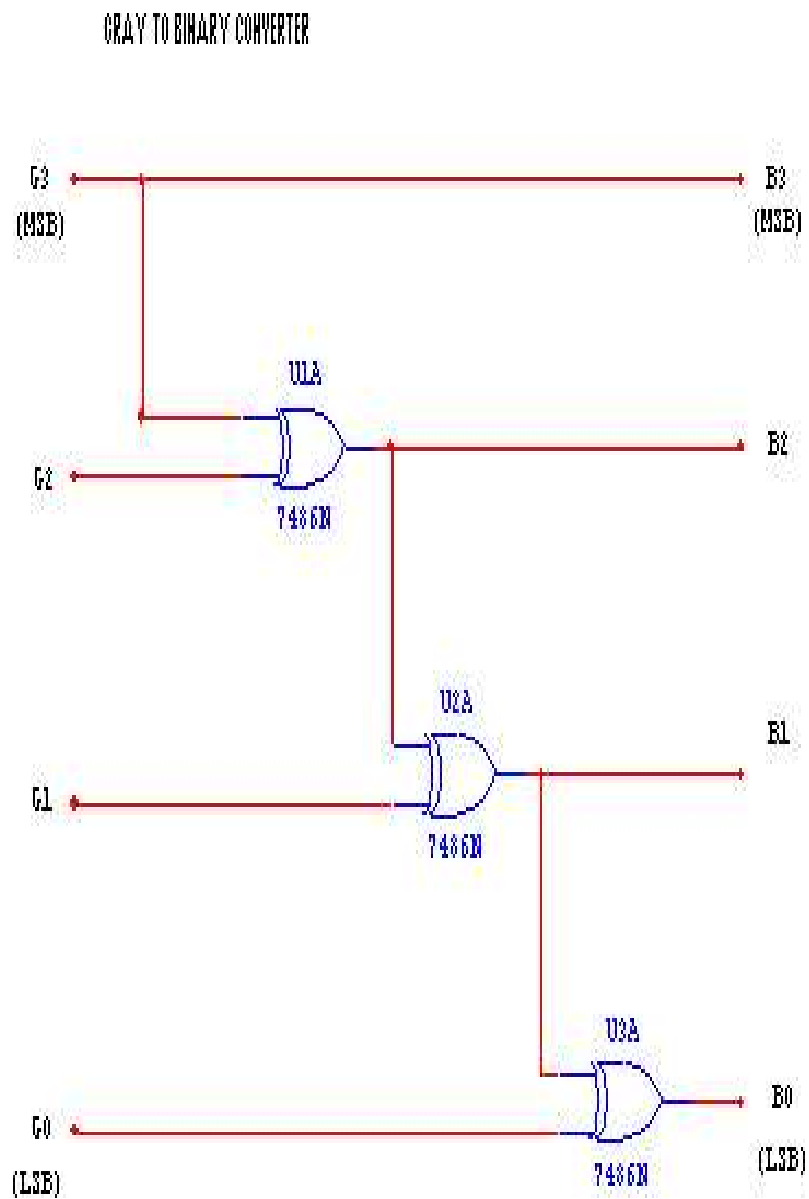


Fig. 7 Logical Circuit Diagram for Gray to Binary Code Conversion



Conclusion:

FAQ's with answers:

Q.1) What is the need of code converters?

There is a wide variety of binary codes used in digital systems. Often it is required to convert from one code to another. For example the input to a digital system may be in natural BCD and output may be 7-segment LEDs. The digital system used may be capable of processing the data in straight binary format. Therefore, the data has to be converted from one type of code to another type for different purpose.

Q.2) What is Gray code?

It is a modified binary code in which a decimal number is represented in binary form in such a way that each Gray-Code number differs from the preceding and the succeeding number by a single bit.

(e.g. for decimal number 5 the equivalent Gray code is 0111 and for 6 it is 0101. These two codes differ by only one bit position i. e. third from the left.) It is non-weighted code.

Q.3) What is the significance of Gray code?

Important feature of Gray code is it exhibits only a single bit change from one code word to the next in sequence. Whereas by using binary code there is a possibility of change of all bits if we move from one number to other in sequence (e.g. binary code for 7 is 0111 and for 8 it is 1000). Therefore it is more useful to use Gray code in some applications than binary code.



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Q.4) What are applications of Gray code?

1. Important feature of Gray code is it exhibits only a single bit change from one code word to the next in sequence. This property is important in many applications such as Shaft encoders where error susceptibility increases with number of bit changes between adjacent numbers in sequence.
 2. It is sometimes convenient to use the Gray code to represent the digital data converted from the analog data (Outputs of ADC).
 3. Gray codes are used in angle-measuring devices in preference to straight forward binary encoding.
 4. Gray codes are widely used in K-map
- Q.5) What are weighted codes and non-weighted codes?

In weighted codes each digit position of number represents a specific weight. The codes 8421, 2421, and 5211 are weighted codes.

Non weighted codes are not assigned with any weight to each digit position i.e. each digit position within the number is not assigned a fixed value. Gray code, Excess-3 code are non-weighted code.

Q.6) Why is Excess-3 code called as self-complementing code?

Excess-3 code is called self-complementing code because 9's complement of a coded number can be obtained by just complementing each bit.

Q.7) What is invalid BCD?

With four bits, sixteen numbers (0000 to 1111) can be represented, but in BCD code only 10 of these are used as decimal numbers have only 10 digits from 0 to 9. The six code combinations (1010 to 1111) are not used and are invalid.